Roadmap. Srednicki §5: scattering amplitudes from time-ordered correlators. Wave packets $a_1^\dagger(t)$ and (5.11)–(5.12) (pp. 1–4) give the LSZ formula (pp. 5–6). pp. 6–10, brief in class, show when it holds: $\langle 0|\varphi|0\rangle = 0$ and $\langle k|\varphi(x)|0\rangle = e^{-ikx}$ are imposed; multiparticle states drop out by themselves. Next, LecNote5–8 (§§6–9) compute the correlators and LecNote9 (§10) applies LSZ; LecNote8 and 12 fix p. 10's Y, Z_φ, Z_m. Supplements: AsymptoticStates, RiemannLebesgue, FeynmanRulesLSZ, QFTHistory (Aside A).

Chapter 5

LSZ Reduction

(Lehmann – Symanzik – Zimmermann)

We can get particle state

$$|k\rangle = a^\dagger(\vec{k})|0\rangle$$

$$a^\dagger(\vec{k}) = -i\int d^3x\, e^{ikx} \overleftrightarrow{\partial_0} \varphi(x)$$

$$a(\vec{k})|0\rangle = 0$$

We require the normalization, such that

$$\langle 0|0\rangle = 1$$

Then the one-particle state will have the

Lorentz invariant normalization

$$\langle k|k'\rangle = \langle 0|a(\vec{k})\, a^\dagger(\vec{k}')|0\rangle \quad (1)$$

$$= (2\pi)^3 2\omega\, \delta^3(\vec{k}' - \vec{k})\langle 0|0\rangle$$

$$= (2\pi)^3 2\omega\, \delta^3(\vec{k}' - \vec{k})$$

Now we can define a time-independent operator that create a particle localized in $\vec{k}$ and near the origin $\vec{x} = 0$.

1 Should read $\langle 0|a(\mathbf{k})\, a^\dagger(\mathbf{k}')|0\rangle$, since $\langle k| = (a^\dagger(\mathbf{k})|0\rangle)^\dagger = \langle 0|a(\mathbf{k})$. As written it vanishes: $a(\mathbf{k}')|0\rangle = 0$.

(Recall "minimal wave packet" in QM free particle)

$$a_1^\dagger = \int d^3k \; f_1(\vec{k})\, a^\dagger(\vec{k})$$

where $f_1(\vec{k}) \propto e^{-(\vec{k}-\vec{k}_1^{(0)})^2/4\sigma^2}$

σ is the spread in momentum space.

If we time evolve this wave packet (in Schrödinger picture), ~~both~~ the ~~two~~ localized particle in $\vec{x}=0$ will spread out at $t \to \pm\infty$

If we consider two particles

$a_1^\dagger a_2^\dagger |0\rangle$, at $t \to -\infty$, these two particles are both spreaded out the space, but as long as $\vec{k}_1 \neq \vec{k}_2$, they are separated. (wildly)

Now [Assume] it still works for interacting theory.

(we will be back to this point soonish.)

$a^\dagger(\vec{k})$ can no longer be time-independent.
(as they will meet and interact).

A suitable initial state is still $(\vec{k}_1 \neq \vec{k}_2)$

$$|i\rangle = \lim_{t\to-\infty} a_1^\dagger(t)\, a_2^\dagger(t)\,|0\rangle$$

but proper normalization, we can make $\langle i|i\rangle = 1$

Similarly, final states

$$|f\rangle = \lim_{t\to+\infty} a_{1'}^{\dagger}(t)\, a_{2'}^{\dagger}(t)\,|0\rangle$$

$(\vec{k}_{1'} \neq \vec{k}_{2'})$ (1)

1 Also take every final momentum different from every initial one, $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$. Otherwise $\langle f|i\rangle$ also contains the no-scattering part of $S = 1 + iT$: terms built from the packet overlaps $O_{i'j} \sim e^{-(\mathbf{k}_{i'}-\mathbf{k}_j)^2/8\sigma^2}$, which the step on p. 5 drops. (Free theory: p. 5 gives 0, while $\langle f|i\rangle = O_{1'1}O_{2'2} + O_{1'2}O_{2'1}$.)

Now the scattering amplitude is given by

$\langle f|i\rangle$.

and we need to find a more useful expression to evaluate $\langle f|i\rangle$.

Note that

$$a_1^\dagger(+\infty) - a_1^\dagger(-\infty) = \int_{-\infty}^{\infty} dt\ \partial_0 a_1^\dagger(t)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x\, \partial_0 \left(e^{ikx} \overleftrightarrow{\partial_0} \varphi(x) \right)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x\, e^{ikx} (\partial_0^2 + \omega^2) \varphi(x)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x\, e^{ikx} (\partial_0^2 + \vec{k}^2 + m^2) \varphi(x)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x \left(e^{ikx} (\partial_0^2 + \vec{\nabla}^2 + m^2) \varphi(x) + (\vec{\nabla}^2 e^{ikx}) \varphi(x) \right)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x \left(e^{ikx} (\partial_0^2 - \vec{\nabla}^2 + m^2) \varphi(x) + \text{surface terms (dropped)} \right)$$

$$= -i \int d^3k\, f_1(\vec{k}) \int d^4x\, e^{ikx} (-\partial^2 + m^2) \varphi(x)$$

In a free field theory $(-\partial^2 + m^2)\varphi(x) = 0$
as $\varphi(x)$ satisfies the Klein-Gordon Equation.

In an interacting theory, (say) $\mathcal{L}_I = \frac{1}{6} g \varphi^3$

how the E.o.M. gives

$$(-\partial^2 + m^2)\varphi(x) = \frac{1}{2} g \varphi^2$$

then $a_1^\dagger(-\infty) = a_1^\dagger(+\infty) + i \int d^3k f_1(\vec{k}) \int d^4x\, e^{ikx} (\partial^2 + m^2) \varphi(x)$. ... (5.11)

h.c. of the above yields.

$$a_1(+\infty) = a_1(-\infty) + i \int d^3k\, f_1(\vec{k}_1) \int d^4x\, e^{-ikx} (\partial^2 + m^2) \varphi(x) \quad \text{... (5.12)}$$ [1]

1 Should read $f_1(\mathbf{k})$, as in (5.11): the h.c. gives $f_1^*(\mathbf{k})$, and f_1 is real (p. 2). The operator in both lines is $(-\partial^2 + m^2)$; its minus is cramped against the parenthesis.

~~so~~ back to

$$\langle f|i\rangle = \langle 0|\, a_{1,1}(+\infty)\, a_{2,1}(+\infty)\, a_2^\dagger(-\infty)\, a_1^\dagger(-\infty)\,|0\rangle$$

we introduce a "Time order" T.

$$\langle f|i\rangle = \langle 0|\, T\, a_{1,1}(+\infty)\, a_{2,1}(+\infty)\, a_2^\dagger(-\infty)\, a_1^\dagger(-\infty)\,|0\rangle$$

T means anything to the right of it are ordered (rearranged) in time, regardless of how written.
(Left is later time and right is earlier time).

With (5.11) and (5.12)

$$\langle f|i\rangle = \langle 0|\, T\{(a_{1,1}(-\infty) + i\int \quad)(a_{2,1}(-\infty) + i\int \quad)$$

~~$a_1^\dagger(-\infty)\, a_2^\dagger(-\infty)$~~

$$(a_1^\dagger(+\infty) + i\int \quad)(a_2^\dagger(+\infty) + i\int \quad)\}|0\rangle$$

= (any $a, a^\dagger$ in above is ~~0~~ zero when acting on $|0\rangle$, or $\langle 0|$) [1]

$$= i^{2+2} \not{*} \int d^4x_1' d^4x_2' d^4x_1 d^4x_2$$

$$e^{-ik_1'x_1'}\, e^{-ik_2'x_2'}\, e^{ik_1x_1}\, e^{ik_2x_2}$$

$$(-\partial_1^2 + m^2)(-\partial_{1'}^2 + m^2)(-\partial_2^2 + m^2)(-\partial_{2'}^2 + m^2)$$

$$\langle 0|\, T\varphi(x_1)\varphi(x_2)\varphi(x_1')\varphi(x_2')\,|0\rangle$$

1 $a^\dagger(+\infty)$ is latest, far left: $\langle 0|a^\dagger(+\infty) = 0$, the out-vacuum. $a(-\infty)$ is earliest, far right: $a(-\infty)|0\rangle = 0$, the in-vacuum (p. 6). Exact up to packet-overlap terms (p. 3).

$$\langle f|i\rangle_n = i^{n+n'} \int d^4x_1\, e^{ik_1x_1}(-\partial_1^2+m^2)\cdots \int d^4x'_1\, e^{-ik'_1x'_1}(-\partial_{1'}^2+m^2)\cdots$$ (1)

$$\times \langle 0|T\,\underbrace{\varphi(x_1)\cdots}_{n}\ \underbrace{\varphi(x'_1)\cdots}_{n'}|0\rangle$$

this is the **LSZ reduction** formula.

one of the key equations of QFT.

Important additional considerations (Canonical quantization makes these transparent)

But here we are assuming that $\langle f|$ and $|i\rangle$ are described by free particles while $\langle f|i\rangle \neq 0$ due to interaction. **Is it a valid assumption?**

The rest is only broadly mentioned during class

(probably skip this part if time is insufficient) following

$\langle 0|0\rangle = 1$ should still be valid (2)

~~$a^\dagger|0\rangle = 0$?~~

$$\varphi(x) = \int d^3x\, e^{-ikx} a(\vec{k}) + a^{-ikx} a^\dagger(\vec{k})$$ (3)

$\langle 0|\varphi(x)|0\rangle$ should be zero

1 Plane waves (here and in p. 5's last lines): $\sigma \to 0$, $f_1(\mathbf{k}) \to \delta^3(\mathbf{k}-\mathbf{k}_1)$ turns every $\int d^3k\, f_1(\mathbf{k})\, e^{ikx_1}$ into $e^{ik_1x_1}$, k_1 on shell. Plane-wave states are δ-normalized, so $\langle i|i\rangle = 1$ is given up: $|\langle f|i\rangle|^2$ will carry $\delta^4(0)$ (LecNote10, §11).

2 So $|0\rangle$ must be both the in-vacuum and the out-vacuum: $\langle 0_{\text{out}}|0_{\text{in}}\rangle = 1$. The vacuum is stable, the unique ground state of a time-independent H, with $P^\mu|0\rangle = 0$. This fails in a time-dependent external field, which creates particles.

3 Free field: $\varphi(x) = \int \widetilde{dk}\,[a(\mathbf{k})e^{ikx} + a^\dagger(\mathbf{k})e^{-ikx}]$: the integral is over $\mathbf{k}$.

$$\varphi(x) = \exp(-iP^\mu x_\mu)\,\varphi(0)\,\exp(+iP^\mu x_\mu).$$

$\langle 0|\varphi(x)|0\rangle = \langle 0|\varphi(0)|0\rangle$ as $P^\mu|0\rangle = 0$.

$= \text{const}$ (L.I.).

~~out~~ we want $a^\dagger(\vec{k})|0\rangle$ to create a single particle state, not a linear combination of 0, single and multiple particle state at $t = \pm\infty$. [1]

To satisfy $\langle 0|\varphi(x)|0\rangle = 0$, if we have $\langle 0|\varphi(x)|0\rangle = v$

we redefine $\varphi(x) \to \varphi(x) - v$.

So the new field $\varphi(x)$ has

$\langle 0|\varphi(x)|0\rangle = 0$. ~~as a scalar, we can shift it freely.~~

[1] Needed only as $t \to \pm\infty$, but the limit cannot enforce two of them: the vacuum and one-particle parts of $a_1^\dagger(t)|0\rangle$ never decay (p. 10). So $v = 0$ and $\langle p|\varphi(0)|0\rangle = 1$ are set by hand. P^μ annihilates bra and ket, so v does not depend on x: $v = 0$ at $\pm\infty$ means $v = 0$ everywhere.

Then $\underbrace{\langle p|}_{\text{a one-particle state.}}\varphi(x)|0\rangle = \langle p|e^{-iPx}\varphi(0)e^{+iPx}|0\rangle$

$= e^{-ipx}\underbrace{\langle p|\varphi(0)|0\rangle}_{\text{this is a number.}}$

we want $\langle p|\varphi(x)|0\rangle = 1$ but ~~they~~ it may not, it is a number depends on L.I. function of p^μ, which is $p^2 = -m^2$ and other parameters. ($x^\mu = 0$ here).

for $\langle p|\varphi(0)|0\rangle = Z$, we can again [1]

rescale $\varphi(0) \to \varphi(0)/Z$.

(renormalize).

1 Here Z is the matrix element itself. Elsewhere (e.g. Peskin & Schroeder) this matrix element is called $\sqrt{Z}$, with Z the residue of the propagator pole. That residue is the Z_φ on p. 10, so $Z_\varphi = Z^2$ in this page's notation.

so we have so far

$$\langle p|\varphi(0)|0\rangle = 1, \quad \langle 0|\varphi(0)|0\rangle = 1 \text{ [2]}$$

2 Should read $\langle 0|\varphi(0)|0\rangle = 0$: the shift $\varphi \to \varphi - v$ on p. 7 enforces it.

What about $\langle \underline{p}, n|\varphi(x)|0\rangle$?

n-particle state with total momentum p

we want it to be zero.

$$\langle p, n|\varphi(x)|0\rangle = e^{-ipx}\langle p, n|\varphi(0)|0\rangle$$

$$= e^{-ipx} A_n(p).$$

$A_n(p)$ is a function of 2.1. of various particle momentum product.

Actually, we really only need

$\langle p, n|a^\dagger(\pm\infty)|0\rangle = 0$; such that our assumptions on the initial/final states are reasonable.

Let's be slightly more serious.

We define the $|p,n\rangle$ state more properly.

$$|\psi\rangle = \sum_n \int d^3p\, \psi_n(\vec{p})\, |p,n\rangle.$$

total wave packet.

We want ($t \to \pm\infty$ to be taken later)

$$\langle \psi | a_1^\dagger(t) | 0 \rangle$$

$$= -i \sum_n \int d^3p\, \psi_n^*(\vec{p}) \int d^3k\, f_1(\vec{k}) \int d^3x\, e^{ikx} \overleftrightarrow{\partial_0} \Phi$$

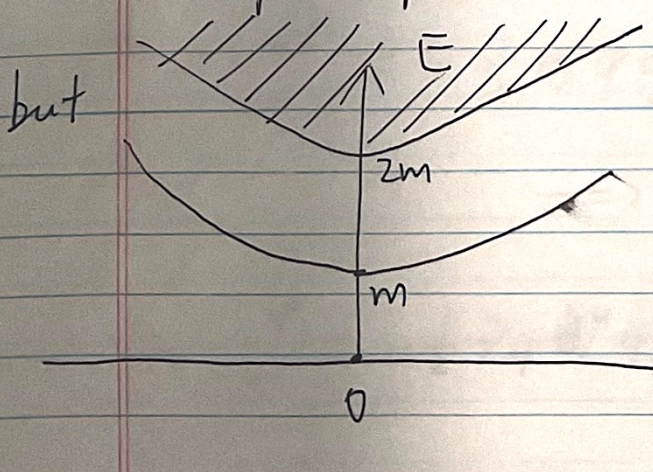
$$\langle p,n | \varphi(x) | 0 \rangle$$

$$= -i \sum_n \int d^3p\, \psi_n^*(\vec{p}) \int d^3k\, f_1(\vec{k}) \int d^3x\, e^{ikx} \overleftrightarrow{\partial_0} e^{-ipx} A_n(\vec{p})$$

$$= \sum_n \int d^3p\, \psi_n^*(\vec{p}) \int d^3k\, f_1(\vec{k}) \int d^3x\, (k^0 + p^0)\, e^{i(k-p)x} A_n(\vec{p})$$

$$= \sum_n \int d^3p\, \psi_n^*(\vec{p})\, f_1(\vec{p})\, (2\pi)^3 (k^0 + p^0)\, e^{-i(k^0 - p^0)t} A_n(\vec{p}) \quad (1) \qquad (5.23)$$

here $p^0 = (\vec{p}^2 + M^2)^{1/2}$, $k^0 = (\vec{p}^2 + m^2)^{1/2}$



we know then

$M \geq 2m > m$

thus. 5.23 osillates fast at $t \to \pm\infty$.

1 Sort by the mass M of the intermediate state. Vacuum: phase e^{-imt}, never decays, so $v = 0$. $M = m$: phase $\equiv 1$, the one-particle packet at every t. $M \geq 2m$: oscillates, $\to 0$. For $a_1(t)|0\rangle$ the one-particle part vanishes identically, so $a_1(\pm\infty)|0\rangle = 0$ (p. 5).

by Riemann-Lebesgue lemma,
the right-hand side vanishes as $t \to \pm\infty$

Physically: one particle wave packet $a^\dagger|0\rangle$ [1]
spread out differently than
$|p,n\rangle$
~~there~~ they are unlikely to overlap via infinite propagation
(at least by constructing one-particle wave packet
carefully).

LSZ formula is good so long as
$\langle 0|\varphi(x)|0\rangle = 0$ and $\langle k|\varphi(x)|0\rangle = e^{-ikx}$.

this means shift & rescale an original Lagrangian to
a new one. e.g.

$$\mathcal{L} = -\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2 + \frac{1}{6}g\varphi^3$$

$\Longrightarrow$ (Will ~~finally~~ formally introduce renormalization later).

$$\mathcal{L} = -\frac{1}{2}Z_\varphi\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}Z_m m^2\varphi^2 + \frac{1}{6}gZ_g\varphi^3 + Y\varphi.$$

we can identify $Z_m m^2/Z_\varphi$ with the physical mass ~~exact~~ and other [2] "physically"

1 Why only $n \geq 2$ decays: $\langle s|a_1^\dagger(t)|0\rangle \propto e^{iK_s t}$, $K = H - \sqrt{\mathbf{P}^2 + m^2}$. Averaged over a packet this decays only if K varies: not for the vacuum ($K = -m$) or one particle ($K \equiv 0$); yes for $n \geq 2$, whose group velocity differs. Same reason the m in $a^\dagger$ must be the physical mass.

2 Backwards: Z_m is chosen so that m itself is the physical mass (the exact propagator's pole at $k^2 = -m^2$; LecNote12). $Z_m m^2/Z_\varphi$ is the bare mass^2, the coefficient for the unrescaled field (LecNote15 p. 13: m_0^2).